

Hook. Southeast Asia's grid is entering a hard-constraint era: electricity demand is moving from 1,300+ TWh today toward 2,000+ TWh by 2035, while renewables are expected to supply more than half of new demand growth. That creates more weather-driven variability, more congestion, and tighter real-time operation. **QuanWatt** turns renewable volatility into actionable grid decisions through a quantum-ready analytics layer for faster power flow, state estimation, contingency screening, transient analysis, and dispatch optimization.

1. Why now: grid analytics becomes the bottleneck

Fast demand growth from cooling, electrification, data centers, and industrial load is colliding with variable solar and wind. Vietnam is the strongest first wedge: it leads Southeast Asia's renewable power market, and its updated 2030 power plan targets **500.4–557.8 billion kWh** of commercial electricity and **89,655–99,934 MW** of peak capacity. For utilities, this means tighter operations, more congestion management, and heavier reliance on simulation and optimization to keep reliability intact.

2. Product: one platform, four utility workflows

QuanWatt is a hybrid quantum-classical grid analytics platform built around the repeated operational kernel behind power-system tools: **large sparse linear solves**. Improving this kernel improves the whole utility toolchain.

- **QPF - Quantum Power Flow:** faster power-flow calculations for high-volume contingency screening and renewable integration studies.
- **QSE - Quantum State Estimation:** faster, more robust real-time power grids visibility from SCADA/PMU measurements.
- **QEMTP - Quantum Transient Analysis:** accelerated time-domain simulation for protection studies and inverter-heavy grid dynamics.
- **QOPF - Quantum Optimal Power Flow:** optimization support for dispatch, congestion management, and market operations for the power systems.

3. Deep-tech edge

- **NISQ path today:** Variational Quantum Linear Solver (VQLS) on simulators or near-term devices, wrapped with our innovative techniques in classical preconditioning, encoding reduction, and measurement schemes to make the quantum computers realistic in the near-term.
- **FTQC path tomorrow:** Our HHL-ready architecture for future early and far fault-tolerant quantum computers, where large sparse systems are the long-term advantage zone.
- **Moat: readout efficiency.** The key blocker for practical quantum linear solvers is measurement/readout overhead. QuanWatt's proposed **Parity Check** method targets the extraction of real coefficient-state information in $O(1)$ settings, partially reducing the readout bottleneck that can erase

speedups, making the NISQ application practical, and paving the way for the FTQC.

- **System integration:** classical preconditioning plus machine-learning-assisted encoding reduction to lower data-loading and circuit-structure complexity.

4. Validation and hackathon demo

The team has built an end-to-end simulation workflow with **Qulacs**, a high-performance Python/C++ quantum-circuit simulator, enabling rapid iteration on VQLS/HHL designs. Initial benchmarks compare the quantum-linear-solver workflow against classical baselines such as NumPy/SciPy. For demo day, QuanWatt can show a renewable-fluctuation scenario on a test feeder with outputs that utility operators understand: voltage-risk flags, overloaded-line alerts, ranked contingencies, and recommended corrective actions.

5. Pilot wedge and go-to-market

- 0–8 mo.** Select one high-renewable Vietnamese town; partner with a university and local utility; validate on real topology and operating data.
- 9–20 mo.** Deploy a low-risk control-center pilot; benchmark against classical workflows; quantify response time, congestion, loss, and stability gains.
- 21+ mo.** Convert pilot results into the first commercial contract; publish a utility case study; replicate across Vietnam, then ASEAN.

6. Why our product matters in the Hackathon

QuanWatt is not a generic quantum pitch. It connects a regional infrastructure pain point to a focused quantum kernel, a defensible algorithmic innovation, and a demo path that can become a utility pilot. The project gives the hackathon a credible quantum-for-energy showcase with climate relevance, Southeast Asia fit, and commercialization potential.

7. Ask

Mentorship on quantum research and power-system validation; simulator or hardware credits; introductions to Vietnam/ASEAN grid-data partners; and support defining pilot metrics that would convince a utility buyer.

Closing thesis: QuanWatt is a practical bridge from NISQ experimentation to real grid value: prove the linear-solver kernel now, deploy it in a narrow utility workflow, and let the advantage compound as quantum hardware matures.

